

VI NGUYEN

Software Engineer Intern

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CAREER OBJECTIVE

Final-year Information Technology student at PTIT HCM, actively building full-stack web applications through personal and team projects. Looking for a Software Engineer Intern position where I can contribute, learn from real-world codebases, and grow alongside an experienced engineering team.

EDUCATION

Posts and Telecommunications Institute of Technology – HCM Campus (PTIT HCM)

2022 – 2027

Bachelor of Information Technology / GPA: 3.22 / 4.0

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript

Backend: NestJS, Spring Boot, Express.js, REST API, Microservices, Apache Kafka, Socket.IO

Frontend: Next.js, React, Ant Design, Tailwind CSS

Database & Storage: PostgreSQL, MySQL, Redis, TypeORM

DevOps & Tools: Docker, CI/CD, Git/GitHub, Swagger, Linux, AI-Assisted Dev (Cursor, Claude Agent, CLI, MCP)

PROJECTS

StudyMate — Learning Management System (LMS)

Full-Stack Developer / *Personal Project* / *Tech:* NestJS, Next.js 15, React 19, TypeScript, PostgreSQL, TypeORM, Redis, Clerk, Cloudinary / [GitHub \(BE\)](#) / [GitHub \(FE\)](#)

- Built a full-stack e-learning platform supporting 4 user roles (Student, Instructor, Staff, Admin) with RBAC, Clerk authentication, and Svx Webhooks for user sync.
- Designed PostgreSQL schema covering course structures, quizzes, lesson progress, and multi-level categories; developed e-commerce features including cart, orders, discount coupons, and instructor payout.
- Integrated Cloudinary for media storage and YouTube Data API v3 for video delivery; applied rate limiting, DTO validation, and Helmet for production hardening.
- Leveraged AI Agents (Claude Code CLI, Cursor) and Model Context Protocol (MCP) to accelerate system design, automate boilerplate code generation, and optimize complex database workflows.

Slack Clone — Real-Time Messaging Platform

Full-Stack Developer / *Personal Project* / *Tech:* NestJS, Next.js, TypeScript, PostgreSQL, Redis, Apache Kafka, Socket.IO, Docker / [GitHub \(BE\)](#) / [GitHub \(FE\)](#)

- Architected a microservices system with 3 independent NestJS services (Identity, Workspace, Chat) each with its own PostgreSQL database; unified traffic through a custom API Gateway.
- Implemented JWT + Google OAuth 2.0 SSO with Redis token blacklisting; built real-time messaging via Socket.IO with Redis Adapter for horizontal WebSocket scaling.
- Integrated Apache Kafka for asynchronous event-driven communication between decoupled services.

VietSneaker — Second-Hand Shoe Marketplace

Backend Developer / *Team Project* / *Tech:* Java, Spring Boot, Spring Security, MySQL, Redis, Next.js, TypeScript, OAuth2, JWT / [GitHub](#)

- Built RESTful APIs with Spring Boot covering product catalog, search/filtering, cart, checkout, and order management; implemented Spring Security with JWT, OAuth2, and RBAC across 4 roles.
- Integrated AI-based recommendation engine via OpenRouter/DeepSeek APIs; built admin dashboard with inventory management, supplier tracking, and revenue analytics.

HCMC Metro Smart Management System

Full-Stack Developer / *Team Project* / *Tech:* React 19, TypeScript, Vite, Ant Design, Recharts, Leaflet, Node.js, Express.js, PostgreSQL, TensorFlow.js, PayOS / [GitHub \(BE\)](#) / [GitHub \(FE\)](#)

- Built an Express.js backend covering authentication (including biometric via TensorFlow.js), e-ticketing, scanner validation, promotions, and admin operations; documented APIs with Swagger and automated ticket expiration via node-cron.
- Integrated PayOS payment gateway for transaction processing and implemented multi-language support (i18next) for local and international users.
- Developed the React 19 admin dashboard with Ant Design, featuring real-time metro analytics (Recharts), interactive station maps (Leaflet), and smooth UX transitions (Framer Motion).

LANGUAGES

TOEIC: 840/990

English: Upper-Intermediate

Able to read technical documentation, API references, and engineering articles; comfortable with practical communication in English.